

Field Theory

Exercise 1. Give an example of an infinite field whose characteristic is not 0.

Solution 1. There are many possible answers, perhaps the easiest is $\mathbb{F}_p(t)$, the field of rational functions over $\mathbb{F}_p$ (the field of fractions of $\mathbb{F}_p[t]$).

Exercise 2.

- (a) Suppose F_1, F_2 are fields and $\phi: F_1 \rightarrow F_2$ is a ring homomorphism. Show that ϕ is either injective or is constantly equal to 0.
- (b) Suppose F is a field of characteristic p . Prove that F has a subfield isomorphic to $\mathbb{Q}$ if $p = 0$, or to $\mathbb{F}_p$ if $p > 0$.
- (c) Find an example of a field F and an injective homomorphism $\phi: F \rightarrow F$ which is not surjective.

Solution 2.

- (a) You can do this by direct calculation. The fast way is that the kernel of ϕ is an ideal of F_1 , but there are only two ideals in a field: (0) and (1) .
- (b) There is a map $\mathbb{Z} \rightarrow F$ which sends 1 to 1. If F has positive characteristic, the image of this map is isomorphic to $\mathbb{F}_p$, and otherwise the image of this map is isomorphic to $\mathbb{Z}$. In the latter case, since F is a field it furthermore contains the field of fractions of $\mathbb{Z}$.
- (c) One example is the map $\mathbb{F}_p(t) \rightarrow \mathbb{F}_p(t)$ which sends $t \mapsto t^2$.

Exercise 3. If a field K is an extension of a field F , then K can also be thought of as a vector space over F .

- (a) Consider $\mathbb{Q}(\sqrt{2})$ as a $\mathbb{Q}$ -vector space. Find a basis.
- (b) Consider $\mathbb{Q}(\sqrt[3]{2})$ as a $\mathbb{Q}$ -vector space. Find a basis.
- (c) Suppose $f \in \mathbb{Q}[x]$ is irreducible of degree n , and consider $\mathbb{Q}[x]/(f)$ as a $\mathbb{Q}$ -vector space. Find a basis.

Solution 3.

- (a) One possible basis is given by $\{1, \sqrt{2}\}$.
- (b) One possible basis is given by $\{1, \sqrt[3]{2}, \sqrt[3]{4}\}$.
- (c) One possible basis is given by the images of the powers of x : $\{1, x, \dots, x^{n-1}\}$.

Exercise 4. If K is an extension of F , its **degree** $[K : F]$ is the dimension of K as an F -vector space.

- (a) Suppose $L \supset K \supset F$ is a tower of extensions. Prove that $[L : F] = [L : K][K : F]$ (that is, degrees multiply in towers).
- (b) Suppose $f \in \mathbb{Q}[x]$ is irreducible, and $\alpha \in \mathbb{C}$ is a root of f . What is $[\mathbb{Q}(\alpha) : \mathbb{Q}]$?
- (c) Give examples of distinct elements $\alpha, \beta \in \mathbb{C}$ such that

$$[\mathbb{Q}(\alpha, \beta) : \mathbb{Q}] < [\mathbb{Q}(\alpha) : \mathbb{Q}][\mathbb{Q}(\beta) : \mathbb{Q}].$$

Can you find an example where $\alpha \notin \mathbb{Q}(\beta)$ and $\beta \notin \mathbb{Q}(\alpha)$?

Solution 4.

- (a) Choose a basis $k_1, \dots, k_m$ for K/F and a basis $\ell_1, \dots, \ell_n$ for L/K . We will show that the set of products $k_i \ell_j$ forms a basis for L/F . Certainly it spans, since for an arbitrary $x \in L$, we can write $x = y_1 \ell_1 + \dots + y_n \ell_n$, where each $y_j \in K$. But then, expanding out the y_j as $\sum_i a_{ij} k_i$ with $a_{ij} \in F$, we get that $x = \sum_{i,j} a_{ij} k_i \ell_j$. On the other hand, suppose that some $\sum_{i,j} a_{ij} k_i \ell_j = 0$. Write $y_j = \sum_i a_{ij} k_i$, then our original sum becomes $\sum_j y_j \ell_j$. Thus, all the y_j must be 0 since the ℓ_j are linearly independent. But then all the a_{ij} must be 0 since the k_i are linearly independent. Thus, the $k_i \ell_j$ are linearly independent.

(A slick proof of this that Josh Mundinger and Ivan figured out is to use tensor products. Notice that $L \otimes_F L$ is an L -vector space of dimension $[L : F]$, so we will be done if we can write this as a tensor product of two L -vector spaces of dimensions $[L : K]$ and $[K : F]$. The most immediate choices would be $L \otimes_K L$ for the former, and $L \otimes_F K$ for the latter. Indeed,

$$\begin{aligned} L \otimes_F L &= (L \otimes_L L) \otimes_F (K \otimes_K L) \\ &= (L \otimes_L (L \otimes_F K)) \otimes_K L \\ &= L \otimes_K (L \otimes_L (L \otimes_F K)) \\ &= (L \otimes_K L) \otimes_L (L \otimes_F K), \end{aligned}$$

as desired. Here, we have used that $M \otimes_R R = M$ and that the tensor product is commutative and associative.)

- (b) $[\mathbb{Q}(\alpha) : \mathbb{Q}] = \deg(f)$
- (c) As the second question gives away, you could choose it so that say $\alpha = \sqrt{2}$ and $\beta = \sqrt[4]{2}$. To get an example for the more difficult extra question, you need $\mathbb{Q}(\alpha) \cap \mathbb{Q}(\beta)$ to be a nontrivial extension of $\mathbb{Q}$ properly contained in both. The smallest way to have this happen would be to have $\mathbb{Q}(\alpha) \cap \mathbb{Q}(\beta) = \mathbb{Q}(\sqrt{d})$, and then $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{d}, \sqrt{a})$ and $\mathbb{Q}(\beta) = \mathbb{Q}(\sqrt{d}, \sqrt{b})$ where a/b is not square in $\mathbb{Q}(\sqrt{d})$. For any choice of a, b, d , this can always be done by the Primitive Element Theorem. As a specific example, we can have $\mathbb{Q}(\alpha) \cap \mathbb{Q}(\beta) = \mathbb{Q}(\sqrt{2})$, and choose $\alpha = \sqrt[4]{2}$ and $\beta = \zeta_8 = \frac{1+i}{\sqrt{2}}$. Then $\mathbb{Q}(\alpha, \beta) = \mathbb{Q}(\zeta_{16})$, which is degree 8 over $\mathbb{Q}$, while both $\mathbb{Q}(\alpha)$ and $\mathbb{Q}(\beta)$ are degree 4.

Exercise 5. A polynomial $f \in \mathbb{Q}[x]$ is said to be **separable** if it has distinct roots in $\mathbb{C}$, and **inseparable** otherwise.

- (a) Give an example of $f \in \mathbb{Q}[x]$ which is inseparable and reducible.
- (b) Prove that f is separable if and only if $\gcd(f, f') = 1$.
- (c) Is it possible for $f \in \mathbb{Q}[x]$ to be inseparable *and* irreducible?

Solution 5.

- (a) An example is given by any polynomial with a double root, e.g. $(x - 1)^2$.
- (b) Let us establish the contrapositive: f has a multiple root iff $\gcd(f, f') \neq 1$. Suppose first that f has a multiple root in $\mathbb{C}$, so over the complex numbers $f = (x - a)^2 g$ for some polynomial g . Then, by the product rule, $f' = 2(x - a)g + (x - a)^2 g' = (x - a)(2g + (x - a)g')$, so $(x - a) \mid \gcd(f, f')$. On the other hand, suppose that all the roots of f are simple, then for each root $a \in \mathbb{C}$ we can write $f = (x - a)h$, where $h(a) \neq 0$. Then we have $f' = h + (x - a)h'$, which does not vanish at a . So, $\gcd(f, f') = 1$.

Exercise 6. Here are two facts you should know:

- If K is a field of characteristic p , then for any $a, b \in K$, $(a + b)^p = a^p + b^p$ by the Binomial Theorem. This means the function $\text{Fr}: K \rightarrow K$ defined by $\text{Fr}(x) = x^p$ is a ring homomorphism. (And it is always injective by Exercise 2.)
 - (Fermat's Little Theorem) In $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$, every element satisfies $x^p - x = 0$. So, $\text{Fr} = \text{id}$, and in particular is surjective.
- (a) Suppose K is a finite field of characteristic p . Prove that Fr is surjective, hence is an isomorphism $K \rightarrow K$.
 - (b) Let $K = \mathbb{F}_3(i)$, where i is a root of the polynomial $x^2 + 1$ (which is irreducible over $\mathbb{F}_3$). Check that, in this case, $\text{Fr} \neq \text{id}$.

Solution 6.

- (a) As noted above, Fr is always injective, so if K is finite then Fr is an injective function between finite sets of the same size, hence must be a bijection.
- (b) We have that $\text{Fr}(i) = i^3 = -i \neq i$, so $\text{Fr} \neq \text{id}$.

Exercise 7. (August 2023 Problem 2) Let p be an odd prime number. Prove that any two fields of order p^2 are isomorphic. (Do not just quote the theorem that there's a unique isomorphism class of fields of any prime power order; your job is to prove a case of that theorem!)

Solution 7. Any field of order p^2 must be a degree 2 extension of $\mathbb{F}_p$, and hence is of the form $\mathbb{F}_p[\sqrt{a}]$ for some non-square $a \in \mathbb{F}_p$, since $p \neq 2$. The multiplicative group $\mathbb{F}_p^\times$ is cyclic, so let $g \in \mathbb{F}_p^\times$ be some generator. Suppose $a \in \mathbb{F}_p^\times$ is nonsquare, and write $a = g^k$ for some odd k . Then we have that $(\sqrt{g^k})^2 = a$, and thus $\sqrt{g^k}$ is a square root of a , so $\sqrt{a} \in \mathbb{F}_p[\sqrt{g}]$. Thus, $\mathbb{F}_p[\sqrt{a}] \subset \mathbb{F}_p[\sqrt{g}]$, but both are degree 2 extensions of $\mathbb{F}_p$, so they must be equal. Hence, every field of order p^2 is $\mathbb{F}_p[\sqrt{g}]$.

(Of course, this result is true also if $p = 2$, and for higher powers than just p^2 . However, imitating the above argument is not a good way to try to prove the general case, a more clever argument is required. You can think for a while about how you might do this, then see the argument in Dummit and Foote.)

Exercise 8. (January 2025 Problem 3) Let F be a field, $K \supset F$ be a finite extension of F , and let a be an element of K .

- (a) Suppose the degree $[K : F]$ is relatively prime to 6. Show that $F(a) = F(a^3)$.
- (b) Suppose the degree $[K : F]$ is not divisible by 3. Either prove that we still have $F(a) = F(a^3)$, or show by example that this is false.

Solution 8.

- (a) We have that $F(a^3) \subset F(a)$, so we have a tower of field extensions:

$$\begin{array}{c} K \\ | \\ F(a) \\ | \\ F(a^3) \\ | \\ F \end{array}$$

Also, a is a root of the polynomial $x^3 - a^3$ over $F(a^3)$, so the degree $[F(a) : F(a^3)]$ is at most 3. On the other hand, the degree of $[F(a) : F(a^3)]$ divides $[K : F]$, which we are told is coprime to 6, so we must have $[F(a) : F(a^3)] = 1$, i.e. $F(a) = F(a^3)$.

- (b) This can be false. For example, let ω be a primitive cube root of unity, and consider the extension $F = \mathbb{Q}$, $K = \mathbb{Q}(\omega)$. Then $[K : F] = 2$, which is coprime to 3, but $\mathbb{Q}(\omega^3) = \mathbb{Q}(1) = \mathbb{Q}$.

Exercise 9. (August 2022 Problem 4) Let p be a prime number, $\mathbb{F}_p$ the finite field with p elements, and $K = \mathbb{F}_p(t)$ the field of rational functions in one variable t over $\mathbb{F}_p$. (So that K is the fraction field of the polynomial ring $\mathbb{F}_p[t]$).

- (a) Show that the splitting field of the polynomial $x^p - t \in K[x]$ over K is inseparable.
- (b) Show that the splitting field of $x^p - t - 1 \in K[x]$ over K is the same as the splitting field of $x^p - t$.
- (c) Show that K has a single degree p inseparable extension.

Solution 9.

- (a) A field extension L/K is said to be *inseparable* if there exists some $\alpha \in L$ whose minimal polynomial is inseparable. In this case, let L be the splitting field of $x^p - t$, and let $\alpha = \sqrt[p]{t}$ be a root of $x^p - t$. The polynomial $x^p - t$ is irreducible over K by Eisenstein's criterion, so this is the minimal polynomial of $\sqrt[p]{t}$. However, $x^p - t$ factors as $(x - \sqrt[p]{t})^p$, so this polynomial is not separable.
- (b) Note that $(\sqrt[p]{t} - 1)^p = (\sqrt[p]{t})^p - 1^p = t - 1$, so $\sqrt[p]{t-1} \in K(\sqrt[p]{t})$ and visa versa, so these are the same extension.
- (c) Let L/K be a degree p inseparable field extension. First, we want to show that $L = K(\alpha)$ for some inseparable element α . Certainly, L has an element α whose minimal polynomial is inseparable. Furthermore, since field degrees are multiplicative, we have $p = [L : K] = [L : K(\alpha)][K(\alpha) : K]$, and moreover $[K(\alpha) : K] \neq 1$ (or else $\alpha \in K$ is separable), so $[K(\alpha) : K] = p$ and thus $K(\alpha) = L$. Now, it is a fact that any inseparable irreducible polynomial is of the form $g(x^p)$ for some irreducible polynomial $g(x)$. Thus, the only inseparable irreducible polynomials of degree p are of the form $x^p - f$ for some $f \in K$. But notice that $f(\sqrt[p]{t})^p = f(\sqrt[p]{t^p}) = f(t)$, so that $\sqrt[p]{f} \in K(\sqrt[p]{t})$. Thus, the only degree p inseparable extension of K is $K(\sqrt[p]{t})$.

Exercise 10. (January 2023 Problem 4) Consider the finite field $\mathbb{F}_p$ where p is a prime number.

- (a) How many monic irreducible polynomials of degree 5 are there over $\mathbb{F}_p$? (Your answer should be a function of p .)
- (b) Let $P(x), Q(x) \in \mathbb{F}_p[x]$ be irreducible polynomials. Give a necessary and sufficient condition for there to exist a third polynomial $R(x) \in \mathbb{F}_p[x]$ such that $Q(x)$ divides $P(R(x))$. (The polynomial $R(x)$ need not be irreducible.)

Solution 10.

- (a) If f is an irreducible polynomial of degree 5, then it has 5 distinct roots in $\mathbb{F}_{p^5}$, which is the unique degree 5 extension of $\mathbb{F}_p$. Distinct irreducible polynomials will have distinct roots. There are p^5 total elements of $\mathbb{F}_{p^5}$, but p of those elements lie in the base field $\mathbb{F}_p$. Thus, the number of degree 5 elements of $\mathbb{F}_{p^5}$ is $p^5 - p$, and since each irreducible polynomial corresponds to 5 distinct roots, we have that the number of irreducible polynomials is $\frac{p^5 - p}{5}$. (Can you independently confirm that for any prime number p , $p^5 - p \equiv 0 \pmod{5}$?)
- (b) Saying that $Q(x) \mid P(R(x))$ is the same as saying $P(R(x)) = 0 \pmod{Q(x)}$. But $\mathbb{F}_p[x]/Q(x) \cong \mathbb{F}_{p^d}$ for $d = \deg Q(x)$, where the isomorphism is given by $x \mapsto \alpha$ for some α a root of $Q(x)$ in $\mathbb{F}_{p^d}$. Then saying $P(R(x)) = 0 \pmod{Q(x)}$ is the same as saying $P(R(\alpha)) = 0$ in $\mathbb{F}_{p^d}$. But then $R(\alpha)$ is simply some element in $\mathbb{F}_{p^d}$, and every element of $\mathbb{F}_{p^d}$ can be written as $R(\alpha)$ for some R , so this is the same as saying P has a root in $\mathbb{F}_{p^d}$, which happens if and only if $\deg P \mid \deg Q$.